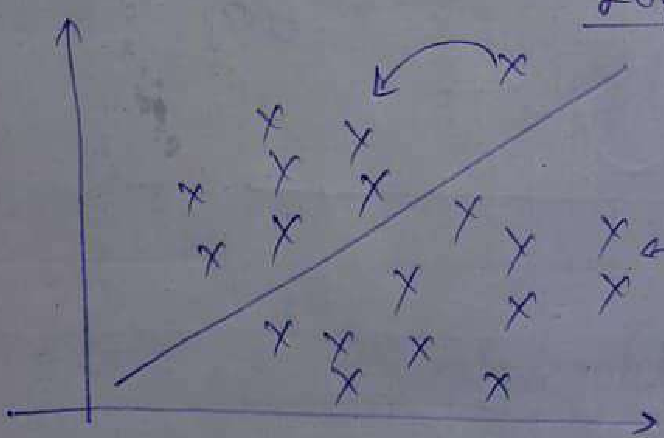
Performance Metrics, Accuracy, Precision, Recall And F-Beta

Topics to be covered

1) Confusion matrix
2) Accuracy
3) Precision
4) Recall
5) F-Beta score



Logistic Regression

R squared

Adjusted R Squared

Dataset

x_1	x_2	y (o/p)	$\hat{y}$ (Pred by model)
-	-	0	1
-	-	1	1
-	-	0	0
-	-	1	1
-	-	1	1
-	-	0	1
-	-	1	0

1. Confusion matrix

Actual values

Predicted values \ Actual values	1	0
1	3	2
0	1	1

Actual

Predicted \ Actual	1	0
1	TP	FP
0	FN	TN

Predicted

② Accuracy $= \dfrac{TP + TN}{TP + FP + FN + TN}$

$$= \frac{3 + 1}{3 + 2 + 1 + 1}$$

$$= \frac{4}{7}$$

④ Dataset

↳ 1000 datapoints → 900 → 1, 100 → 0 (Binary classification)

⇓

90% accuracy

Imbalanced Dataset

③ Precision $= \dfrac{TP}{TP + FP}$ } out of all the ~~predicted~~ actual value how many are correctly predicted

Predicted \ Actual	1	0
1	TP	FP
0	FN	TN

① <u>Recall</u> $= \dfrac{TP}{TP + FN}$ ⇒ out of all the predicted value how many are correctly predicted

<u>Usecase</u> ①

Spam classification

		Actual: spam 1	Actual: Not spam 0
predicted: spam	1	TP	FP
predicted: Not spam	0	FN	TN

Actual ⇓

Mail → spam
Model → spam } Good

Mail → Not spam
Model → spam } blunder

⇓

Precision $= \dfrac{TP}{TP + FP}$

<u>Use case 2</u>

To predict whether person has diabetes or not

Truth → diabetes
Model → doesn't diabetes } Blunder

Truth → diabetes
Model → " } good

Truth → Not diabetes
Model → Diabetes } ⇒ 2^{nd} opinion

⇓

check

	Diabetes	No Diabetes
Diabetes	TP	FP
No Diabetes	FN	TN

↓ ↓↓↓

Usecase of diseases

$$\text{Recall} = \frac{TP}{TP + \boxed{FN}\downarrow\downarrow}$$

Assignment

* tomorrow the stock market will crash or not

Reducing FP ↓ or FN ↓

	Crash	Not crash
Crash	TP	FP
Not crash	FN	TN

↓ J+L

We use **recall** in this case.

Reduce : FN ↓

(5) F - Beta score : $(1+\beta^2)\dfrac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$

(6) If FP & FN are both important

$\beta = 1$

$$\text{F1 score} = (1+1^2)\dfrac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$F_1 \text{ score} = 2 \times \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

Harmonic mean

(2) If FP is more important than FN

$\beta = 0.5$

$$F_{0.5} \text{ score} = (1+0.25) \frac{P * R}{P + R}$$

(3) If FN >> FP

$\beta = 2$

$$\boxed{F_2 \text{ score} = (1+4) \frac{P * R}{P + R}}$$